1. Overall Performance

The XGBoost model achieved an accuracy of 86% and an AUC of 0.81, indicating good overall predictive performance. The AUC value above 0.8 suggests that the model has
a strong ability to distinguish between diabetic and non-diabetic individuals. However, accuracy alone is not sufficient in this case due to the imbalanced nature of the dataset, where non-diabetic cases significantly outnumber diabetic cases.

2. Classification Performance

Class 0 (Non-diabetic):

Precision	Recall
0.89	0.96

Class 1 (Diabetic):

Precision	Recall
0.49	0.27

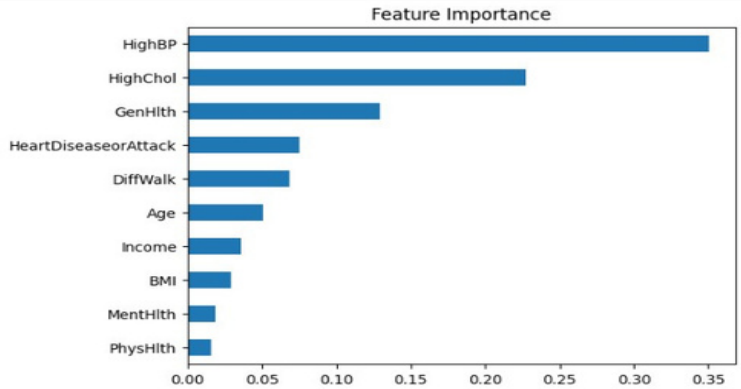
Key Insight

The recall value of 0.27 indicates that the model correctly identifies only 27% of actual diabetic cases, The low recall for the diabetic class is a critical limitation. This means that the model fails to detect a large proportion of true diabetic cases, leading to a high number of false negatives. The model prioritizes minimizing false positives over detecting true diabetic cases. This behavior is undesirable in healthcare applications, where missing a positive case is more critical than generating a false alarm.

3. Feature Importance Analysis

Feature importance analysis using XGBoost:

- High Blood Pressure (HighBP)
- High Cholesterol (HighChol)
- General Health (GenHlth)



4.SHAP Analysis

SHAP analysis was used to interpret the contribution of each feature to the model predictions.

The most influentialfeatures identifiedusingSHAPinclude:

- General Health (GenHlth)
- High Blood Pressure (HighBP)
- Age
- BMI

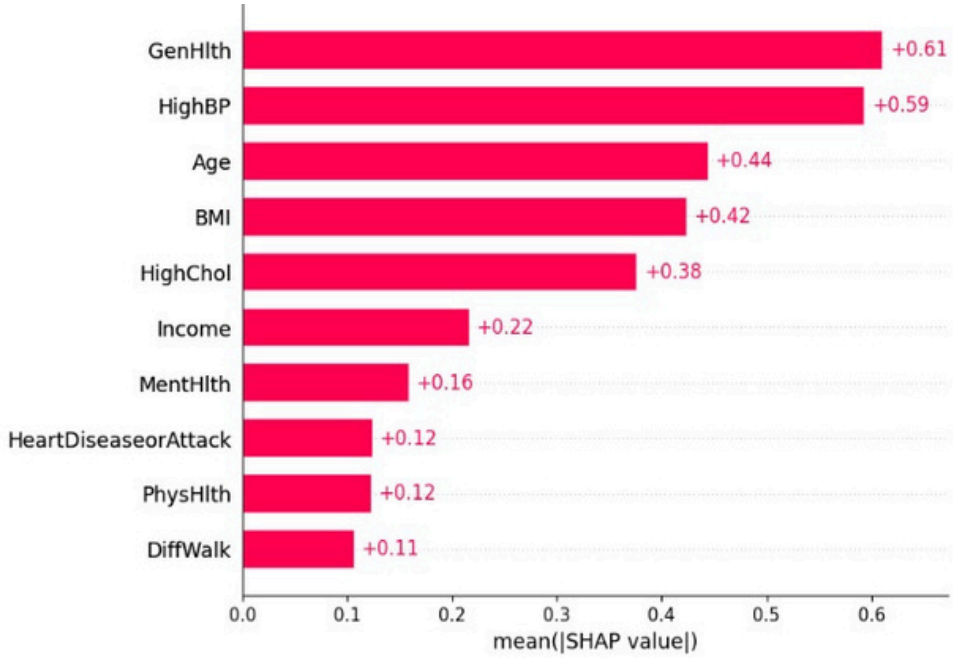


Figure 1:

5.SHAP Detailed Analysis (Beeswarm Plot)

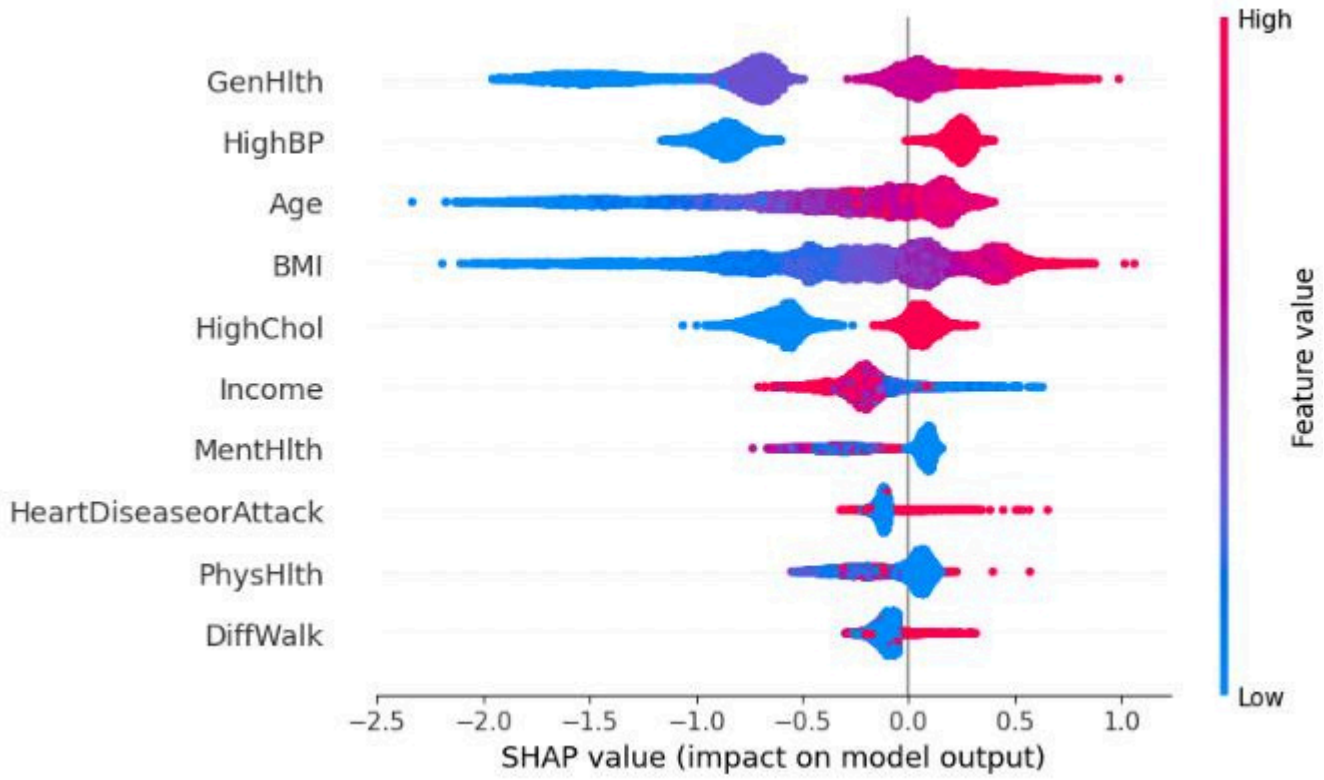
The SHAP beeswarm plot provides detailed insights into how each feature contributes to the model predictions.

A SHAP beeswarm plot was used to provide a more detailed interpretation of the model behavior at the individual prediction level. Unlike the SHAP bar plot, which shows overall feature importance, the beeswarm plot illustrates both the magnitude and direction of each feature's impact across all data samples.

In this plot:

- Each point represents a single observation.
- The horizontal position indicates the impact on the model output.
- The color represents the feature value (low to high).

This allows for deeper insights into how high and low values of each feature influence the prediction differently



6.Feature-Level Insights :

1. General Health (GenHlth)

General health was identified as the most influential feature in the model. Higher values of GenHlth (indicating worse health conditions) are strongly associated with an increased probability of diabetes. Individuals with poorer general health are significantly more likely to be classified as diabetic.

2. High Blood Pressure (HighBP)

HighBP shows a strongpositive impact onpredictions,where individualswith hypertension are consistently pushed toward the diabetic class. This confirms that high blood pressure is a major contributing factor to diabetes risk.

3. Age

Higher age values are associated with increased SHAP values, indicating a higher probability of diabetes. This reflects the well-known relationship between aging and increased diabetes risk.

4. Body Mass Index (BMI)

Higher BMI values contribute significantly to diabetes prediction. This confirms that obesity is a key risk factor captured effectively by the model.

5. High Cholesterol (HighChol)

High cholesterol hasapositive but slightly lower impact compared to blood pressure and age. It plays a secondary but still important role in diabetes prediction.

6. Income

Incomeshows a mixed influence on predictions, indicating variability across individuals. This suggests that socioeconomic factors may indirectly influence diabetes risk.

7. Mental Health (MentHlth)

Mentalhealth hasamoderateand less consistent impact. Its effect is weaker compared to physical health-related features.

8. Heart Disease or Attack

Thisfeature contributespositively to diabetes prediction. It reflects the strong relationship between cardiovascular diseases and diabetes.

9. Physical Health (PhysHlth)

Poorphysical health contributes to increased diabetes risk. This reinforces the importance of overall health condition.

10.Difficulty Walking (DiffWalk)

Difficulty walking hasasmaller but noticeable positive effect. It mayindicate reduced physical activity and higher health risk.

6.Feature-Level Insights :

From the overall SHAP distribution:

- The model heavily relies on general health, blood pressure, age, and BMI as primary predictors.
- Higher feature values generally push predictions toward the diabetic class.
- The model captures complex and non-linear relationships between features.
- Feature impact varies across individuals, indicating personalized prediction behavior.
- The learned patterns are consistent with real-world medical knowledge.

6. Overall Model Behavior:

Themodelperforms wellindistinguishing between diabetic and non-diabetic individuals overall. However, it struggles to detect diabetic cases due to class imbalance.

Conclusion

The XGBoost model achieved strong overall performance with an accuracy of 86% and an AUC of 0.81. However, it shows a significant limitation in detecting diabetic cases, as reflected by the low recall (0.27). The model is biased toward the majority class and tends to miss positive cases. Feature importance and SHAP analysis confirm that key predictors such as general health, blood pressure, age, and BMI are highly influential and aligned with medical knowledge. While the model is reliable overall, improving recall and handling class imbalance remains necessary for practical deployment.